

CSA0609-DESIGN AND ANALYSIS OF ALGORITHMS

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Exp-02: Common Elements Between Two Arrays and Find its Complexity

Aim:

To find the common elements between two arrays using a brute-force approach and analyze its complexity.

Algorithm:

1. Read the size of the first array.
2. Read the elements of the first array.
3. Read the size of the second array.
4. Read the elements of the second array.
5. Compare every element of the first array with every element of the second array.
6. Store the matching elements.
7. Display the common elements and complexity.

Source Code:

```
n = int(input("Enter size of first array: "))
arr1 = list(map(int, input("Enter first array elements: ").split()))
m = int(input("Enter size of second array: "))
arr2 = list(map(int, input("Enter second array elements: ").split()))
common = []
for i in arr1:
    for j in arr2:
        if i == j and i not in common:
            common.append(i)
print("Common Elements:", common)
```

```
print("Time Complexity: O(n*m)")
```

```
print("Space Complexity: O(k)")
```

Output:

```
Enter size of first array: 5
Enter first array elements: 10 20 30 40 50
Enter size of second array: 5
Enter second array elements: 40 50 60 70 80
Common Elements: [40, 50]
Time Complexity: O(n*m)
Space Complexity: O(k)
```

Result:

The common elements between the two arrays were successfully found using the brute-force approach.